

1. Project Summary, Team, Source PSU, Revision & Accepted Requirements

1.1 Project Summary

This project transforms an instructor-approved, closed-case PC ATX Power Supply Unit (PSU) into a safe, multi-output benchtop power supply for laboratory experimentation. The final design provides fixed DC output rails (+3.3V, +5V, +12V, adjustable) along with an added instructor-approved buck-boost module for variable/adjustable DC output voltage.

To adhere strictly to the non-negotiable project safety boundary, the metal source PSU enclosure remains completely closed and unmodified. All power distribution, branch protection (fuses), control switches, indicators, and adjustable converter controls are housed inside a separate, insulated external enclosure connected strictly through the accessible low-voltage output harness.

1.2 Project & Team Information

Course: 091013107 Electronics Circuits Laboratory
Project Title: Project 1 – ATX Bench Power Supply
Delivery Date: Monday 28 September 2026, 13:00–16:00
Project Revision: Rev 1.0 (As-Built Initial Submission)
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1.3 Source PSU Specifications

Manufacturer: PLENTY COMPUTER

- Model No.: ATX500WS
- Enclosure Status: Intact, closed metal case (no physical modification/opening)
- Connector Type: Standard 24-pin ATX Main Motherboard Connector
- AC Input Ratings:
 - 115V ~ / 10A
 - 230V ~ / 6A
- DC Output Ratings & Limits:
 - +3.3V Rail: 22A
 - +5V Rail: 14A (+3.3V & +5V Combined Max: 130W)
 - +12V1 Rail: 13A (Max: 195W)
 - +12V2 Rail: 14A (Max: 168W)
 - -12V Rail: 0.8A (Max: 9.6W)
 - +5Vsb Rail: 2.5A (Max: 12.5W)
 - Combined Combined Output Power (-12V & +5Vsb): 22W
 - Total Maximum Power: 500W
- Stated Minimum Load Requirement: 0 A

1.4 Accepted Project Requirements & Acceptance Criteria

Requirement Item	Minimum Acceptance Criteria	Acceptance Status
Approved Source PSU	Closed-case, intact AC inlet, manufacturer nameplate photograph verified with instructor approval.	PASS
Fixed Rails Output	Available +3.3V, +5V, +12V rails presented on clearly labelled, insulated binding posts.	PASS
Branch Protection	Separate instructor-approved fuse in every accessible non-COM positive output branch, placed before exposed wiring	PASS
Adjustable Output	Approved buck-boost module fed by protected +12V rail, with labelled controls (CV/CC) and external binding posts.	PASS

Enable Control	Insulated maintained switch connecting verified PS_ON (Pin 16) to COM ; strictly no exposed temporary jumpers.	PASS
Power Indication	Clear visual indicators (LEDs) for Standby status (+5VSB) and Main Power active status.	PASS
Mechanical Enclosure	Enclosed low-voltage distribution box, cable strain relief, secured wire harnesses, and protected ventilation.	PASS
Unpowered Safety	Passed all continuity, rail-to-rail short circuit, polarity, and insulation checks prior to AC connection.	PASS
Electrical Performance	Complete measurements recorded for voltage, path drop, ripple, and thermal stability under approved loads.	PASS
Reproducible Docs	Full GitHub documentation including schematics, BOM, calculations, photos, operating procedures, and contributions	PASS